

Retail Sales, Discounts, and Profitability

An exploratory analysis with R | Wenlong Zhang | September 2026

Executive summary

This portfolio project examines where profitability is weakest in Tableau’s Superstore sample data. Across 10,194 order lines and 5,111 distinct orders, total sales were 2,326,534.35 and total profit was 292,296.81, producing an aggregate profit margin of 12.56%. Furniture had the lowest category margin, while every observed discount group at 30% or higher recorded an aggregate loss. These findings identify areas for review; they do not establish an optimal discount policy.

Business questions and data

The analysis asks which product categories have the weakest margins, how profitability differs across discount levels, and whether a curved regression relationship describes line-level margins better than a straight line. The dataset is a retail sample used for learning, not evidence of results achieved for a real employer.

The imported Orders sheet contained 21 original fields. Checks reported no NA values, no fully duplicated rows and no duplicated Row IDs. Multiple lines can belong to the same order, so line counts and distinct order counts are reported separately. Monetary figures below retain the dataset’s units.

Category performance

Category	Sales	Profit	Margin
Technology	839,893.28	146,543.38	17.45%
Office Supplies	731,893.31	126,023.44	17.22%
Furniture	754,747.76	19,730.00	2.61%
Total	2,326,534.35	292,296.81	12.56%

Furniture generated more sales than Office Supplies but substantially less profit. Its margin was 14.83 percentage points below Technology’s. A review should begin with furniture products and their discount, cost and sales mix, rather than treating revenue growth alone as success.

Measurement approach

Category and discount-group margins equal total profit divided by total sales. Regression uses each order line’s Profit / Sales ratio, with each line weighted equally. These answer different questions: the regression outcome is not the overall business margin. The loss-line percentage is the share of lines with negative profit.

Discount patterns and regression

Higher-discount groups generally showed weaker profitability. The chart summarizes observed groups; it does not show the effect of changing a discount on the same product.

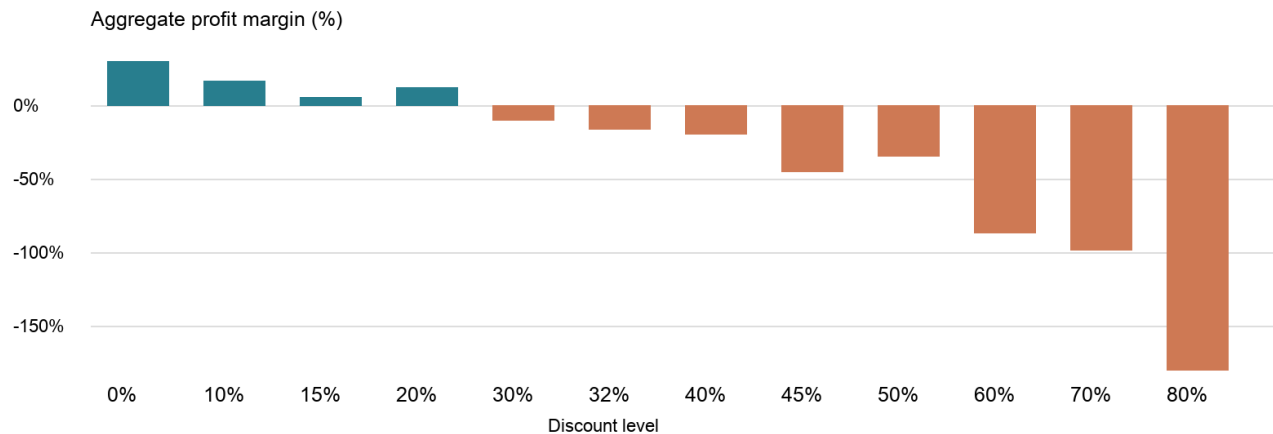


Figure 1. Aggregate margin by discount level, calculated from the saved R summary table.

Discount	Order lines	Group margin	Loss-making lines
0%	4,925	29.56%	0.00%
15%	52	5.15%	32.69%
20%	3,706	11.77%	13.76%
30%	230	-10.06%	91.74%
80%	301	-180.01%	100.00%

Selected groups illustrate both the broad pattern and its exceptions. The 15% group had a lower margin than the 20% group, consistent with differences in product mix. Group sizes ranged from 11 to 4,925 lines. A margin below -100% means the recorded loss exceeded sales revenue for that group; it is not a calculation error by itself.

Comparing regression specifications

Model: line-level profit rate	Adjusted R ²
Discount only	0.7479
Discount + product category	0.7511
Discount + discount squared + product category	0.8470

Adding a squared discount term improved in-sample fit, consistent with a curved relationship. Adjusted R² of 0.8470 is not 84.7% prediction accuracy. The residual plot still showed uneven spread and structure, so the quadratic model remains an exploratory approximation.

In the category-adjusted linear model, a 10-percentage-point higher discount was associated with a 19.47-percentage-point lower line-level margin. This is an association, not a causal effect. The quadratic model allows this slope to vary with discount, so the linear estimate should not be used as a universal forecast.

Interpretation and next actions

Recommended business follow-up

1. Investigate furniture profitability. Break results down by sub-category and product, then examine cost, discount and order size. Identify whether losses are concentrated in a small number of products before making category-wide changes.
2. Review high-discount transactions. The observed groups at 30% and above all lost money in aggregate. Use this as a review flag, not a universal maximum discount. Check inventory clearance, promotional purpose and product economics before deciding whether a promotion should change.
3. Validate any pricing change prospectively. Define profit, units sold and revenue as evaluation metrics, then use a controlled pilot where feasible. This analysis has not estimated demand response or demonstrated that reducing discounts would increase total profit.

Limitations

Observational and sample data. Product mix, geography, customer segment and promotion selection may explain part of the association. Only category was included as an additional control. Findings describe this sample and should not be generalized directly to an operating retailer.

Accounting relationship. Discount, revenue and profit are economically connected. Dividing profit by sales can make margins especially negative at deep discounts. Strong fit therefore does not independently demonstrate a causal mechanism or a successful pricing strategy.

Model validation. Results are from the fitted sample; no holdout prediction assessment was performed. Residual structure and unequal variance remain. Lines may share orders or products, and conventional regression standard errors may not account for those dependencies. This report does not rely on p-values to justify policy changes.

Scope of verification. Report totals were checked against the saved overall, category and discount CSV summaries. Regression statistics and data-quality checks were taken from the supplied R output and screenshots; the original analysis was not rerun for this report.

Project documentation

Tools and methods: R; readxl for Excel import; base R aggregation, visualization and lm() regression. Supporting files: sales_analysis.R, overall_summary.csv, category_summary.csv and discount_summary.csv. The script, source data instructions and report should be kept together so the workflow can be reproduced.

Data source

Tableau, Sample - Superstore, Orders sheet. Public sample dataset used for educational analysis. The results refer to the locally analyzed version; other versions may have different dates or totals.

Source: https://public.tableau.com/app/sample-data/sample_-_superstore.xls

All figures are descriptive findings from the sample. No implemented financial improvement or business impact is claimed.